

Report
Of
**AI-Assisted Marine Risk Intelligence &
Community Reporting Platform**

Submitted
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B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, **NAVNEET KUMAR** do hereby declare that the internship report titled "**AI-Assisted Marine Risk Intelligence & Community Reporting Platform**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I extend my sincere thanks to the entire team at MARS Group for welcoming me as an intern and providing a professional, collaborative environment. Most importantly, I acknowledge the efforts of my team lead, Deepankar Singh Sirari, for providing clear project direction, consistent mentorship, and coordination throughout the internship period. The collaborative environment fostered by our 8-member team made this project a tremendous learning experience that has greatly enriched my academic and professional development.

Signature: _____

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INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP COMPLETION

Reference No.: MARS/INT-CERT/2026/033

Date of Issue: July 31, 2026

This is to certify that

Navneet Kumar

has successfully completed an internship with MARS – Centre for Sustainable Community Development from June 1, 2026 to July 15, 2026, contributing sincerely to the organisation's initiatives in sustainable community development. During this period, the intern demonstrated dedication, a strong learning attitude, and a genuine commitment to the values of sustainability and community service.

Sustainability is not a destination but a continuing journey. May the knowledge and experience gained during this internship inspire you to keep championing sustainable practices, and to carry this commitment forward into every community and career you touch in the years ahead.

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4. PROJECT DESCRIPTION

4.1 Introduction

The AI-Assisted Marine Risk Intelligence & Community Reporting Platform is a geospatial marine intelligence system engineered to fuse satellite sea-surface temperature (SST) data, community-reported coastal incidents, and simulation tools in order to identify at-risk marine zones and support timely environmental response. The platform bridges the gap between raw satellite telemetry and actionable ecological intelligence, making it accessible to NGOs, coastal authorities, and research institutions.

Rising ocean temperatures and deteriorating water chemistry pose existential threats to marine ecosystems worldwide. However, real-time monitoring data is often siloed in technical formats that are inaccessible to decision-makers in the field. This platform addresses that gap by transforming complex oceanographic data into intuitive visualisations and community-driven alerts, enabling faster environmental response.

4.2 Organization Profile

The internship was undertaken with MARS Group, an organisation focused on leveraging data science, artificial intelligence, and geospatial technology for environmental monitoring and risk assessment applications. MARS Group provided the project direction, mentorship, and collaborative team structure required to deliver a production-quality marine intelligence platform.

The project was executed by an 8-member cross-functional team under the leadership of Team Lead Deepankar Singh Sirari. The team comprised specialists in data science, GIS engineering, database engineering, backend development, and frontend development, ensuring comprehensive coverage of all technical domains required for the platform.

4.3 Problem Statement

Marine ecosystems globally are under severe stress due to rising sea-surface temperatures, ocean acidification, declining salinity balance, and increasing CO₂ absorption. Existing monitoring systems either require highly specialised technical

expertise to operate, are geographically limited, or fail to incorporate community-reported ground-truth observations. Consequently, early warning signals are often missed, and environmental response is delayed.

There is a critical need for a platform that can automatically detect SST anomalies against historical baselines, compute multi-parameter risk scores, display risk zones on an interactive GIS map, and empower coastal communities to report pollution or distress events — all within a unified, accessible interface.

4.4 Project Objectives

1. Ingest and process satellite SST data from trusted sources including NOAA, OISST, NCEI, and OceanSODA.
2. Compute SST anomalies relative to a 1991–2020 historical climatology baseline to detect abnormal warming events.
3. Display GIS layers for SST values, anomalies, and classified marine risk zones.
4. Implement a heatwave simulation tool allowing users to model the spread and impact of a marine heat event.
5. Enable community reporting of pollution incidents and harmed coastal areas with automated verification against computed risk zones.
6. Develop a multi-parameter risk framework integrating SST anomaly, pH, salinity, and CO₂ data.
7. Deliver an interactive dashboard for NGOs, coastal authorities, and environmental agencies to support data-driven decision-making.

4.5 Scope of the Project

The platform covers the following functional and technical scope:

1. Geographic coverage of global ocean regions using satellite telemetry at 0.25° to 10° grid resolution.
2. Historical baseline spanning 1991–2020 for climatologically accurate anomaly computation.
3. Integration of four distinct data sources (NOAA, OISST, NCEI, OceanSODA) with heterogeneous formats and conventions.
4. Community reporting module with geolocation tagging and risk-zone cross-verification.

- 5. Geospatial visualisation using QGIS (EPSG:4326/WGS 84) and IDW interpolation techniques.
- 6. Backend REST API deployed via Node.js with a PostgreSQL/PostGIS geospatial database.
- 7. Interactive Streamlit dashboard with Plotly Express for real-time data exploration.
- 8. Comprehensive documentation and reproducible pipeline for future development and maintenance.

4.6 Technologies and Tools Used

Table 4.1 summarises the technologies and tools employed across all modules of the platform.

Domain	Technologies / Tools
Data Processing	Python (Pandas, NumPy, Xarray), NetCDF, CSV
Geospatial Analysis	QGIS (EPSG:4326 / WGS 84), IDW Interpolation, PostGIS
Database	PostgreSQL, pgAdmin, PostGIS, Prisma ORM
Backend	Node.js, Python-to-PostgreSQL Automation Pipeline, REST API
Frontend / Dashboard	Streamlit, Plotly Express, Responsive Web Design
Data Sources	NOAA, OISST, NCEI, OceanSODA
Version Control & Docs	Git, Technical Documentation, Flowcharts

Table 4.1: Technologies and Tools Used

4.7 System Architecture

The platform follows a layered architecture comprising five primary tiers:

Data Ingestion Layer: Satellite SST datasets (NetCDF/CSV) from NOAA, OISST, NCEI, and OceanSODA are downloaded and staged for processing.

Data Processing Layer: Python scripts (Pandas, NumPy, Xarray) clean, validate, transform coordinates, compute SST anomalies, and integrate ocean chemistry parameters.

Geospatial Layer: QGIS with PostGIS performs spatial aggregation, IDW interpolation, risk zone classification, and coordinate-system normalisation (WGS 84).

Backend / API Layer: A Node.js REST API serves processed datasets and community reports to the frontend. A PostgreSQL/PostGIS database with Prisma ORM stores all persistent data.

Presentation Layer: A Streamlit dashboard with Plotly Express provides interactive maps, SST anomaly charts, risk-zone overlays, and community report interfaces.

4.8 Methodology

Step 1 – Data Collection & Research: Conducted extensive research into satellite marine data sources. Identified and compared NOAA, OISST, NCEI, and OceanSODA datasets based on spatial resolution, temporal coverage, and data quality. Downloaded satellite SST data along with ocean chemistry parameters (pH, salinity, CO₂).

Step 2 – Data Cleaning & Quality Assurance: Developed comprehensive cleaning pipelines in Python (Pandas, NumPy). Removed missing values, duplicate records, and invalid coordinates. Standardised column names across heterogeneous sources. Maintained documentation of all cleaning decisions for reproducibility.

Step 3 – Coordinate System Normalisation: Resolved a critical longitude convention mismatch (0–360° satellite convention vs. –180° to +180° standard mapping) by implementing a transformation rule (IF Longitude > 180, THEN Longitude – 360). Cross-verified results against NOAA reference maps.

Step 4 – SST Anomaly Computation: Implemented the SST anomaly formula: Current SST – Historical Mean SST (1991–2020 climatology baseline). Validated outputs against NOAA reference maps. Identified marine heatwave hotspot regions from anomaly outputs.

Step 5 – Multi-Parameter Risk Framework: Investigated interactions between SST anomaly, pH, salinity, and CO₂. Developed a scientifically grounded risk scoring model based on peer-reviewed marine science literature.

Step 6 – Geospatial Optimisation: Reduced dataset resolution from 0.25° to 10° grid through cell-wise averaging to balance spatial accuracy with computational performance. Imported optimised datasets into QGIS with appropriate symbology.

Step 7 – Backend Integration & Testing: Prepared backend-ready exports. Coordinated data handoff with the Database Engineer for PostgreSQL/PostGIS integration. Cross-verified data integrity post-import. Created representative sample datasets for load and performance testing.

Step 8 – Dashboard Development Support: Specified data visualisation requirements for the frontend team, provided sample datasets for dashboard development, and verified visualisation accuracy against source data.

4.9 Expected Outcomes

1. An analysis-ready SST and ocean-chemistry dataset that every other module (GIS, database, backend, dashboard) can reliably consume.
2. Accurate SST anomaly maps identifying marine heatwave hotspots relative to a 1991–2020 climatology baseline.
3. An interactive GIS-based risk map allowing NGOs and coastal authorities to visualise at-risk zones in real time.
4. A community reporting interface enabling field-level pollution and distress event reporting with automated risk-zone verification.
5. A simulation tool allowing decision-makers to model hypothetical marine heatwave scenarios and their spatial propagation.
6. A comprehensive, reproducible data pipeline with full documentation, enabling future maintenance and scalability.
7. Meaningful societal impact by equipping environmental agencies with data-driven tools to respond faster to marine ecological crises.

4.10 Role & Responsibilities

Primary Role: Data Collector & Handler

As the primary Data Collector & Handler, the objective was to progress from theoretical research to a working, GIS-ready data pipeline by collecting reliable datasets, preparing a historical climatology baseline, cleaning and validating all data, calculating SST

anomaly values, optimising datasets spatially, and documenting the complete workflow for scalability and deployment.

Key Responsibilities:

- **Data Collection & Research:** Researched trusted satellite data sources; identified and compared datasets from NOAA, OISST, NCEI, and OceanSODA.
- **Data Cleaning & Validation:** Removed missing values, duplicates, and invalid coordinates; standardised column names across all sources.
- **Analysis & Computation:** Implemented SST anomaly calculations, computed climatology baseline, developed multi-parameter risk framework.
- **Geospatial Optimisation:** Reduced high-resolution datasets to appropriate grid levels; resolved coordinate system issues.
- **Ocean Chemistry Integration:** Processed pH, salinity, and CO₂ parameters; integrated with SST data.
- **Cross-Functional Coordination:** Collaborated with GIS Engineer, Database Engineer, Backend Developer, and Frontend Developer across an 8-member team.
- **Documentation:** Maintained comprehensive pipeline documentation including flowcharts, methodology notes, and handoff documents for all downstream teams.

Key Contributions:

1. Built the foundational, analysis-ready SST and ocean-chemistry dataset that every other platform module depended on.
2. Identified and resolved the critical longitude-convention bug ($0-360^{\circ} \rightarrow -180^{\circ}$ to $+180^{\circ}$), which the GIS engineer then used as the foundation for all coordinate correction work.
3. Designed the multi-parameter risk framework integrating SST anomaly, pH, salinity, and CO₂, providing the backend with a scientifically sound ecological model.
4. Served as a cross-functional bridge between the data science and engineering teams, translating data requirements into implementable formats.
5. Produced comprehensive documentation ensuring the pipeline is maintainable and understandable for future developers.

Challenges & Solutions

Longitude Convention Mismatch: Satellite telemetry recorded longitude on 0–360°, while standard mapping uses –180° to +180°. Resolved by developing a coordinate transformation rule implemented in Python and verified against NOAA reference maps.

Over-Cleaning Risk Near Coastal/Ice Regions: Initial aggressive cleaning removed valid but statistically uncommon readings. Resolved by shifting to incremental cleaning with team review of flagged rows.

Spatial Aggregation Level: High-resolution 0.25° grid was computationally expensive. Resolved by testing multiple aggregation levels (1°, 5°, 10°) and selecting 10° as optimal after pattern validation.

Multi-Parameter Framework Integration: Combining SST anomaly, pH, salinity, and CO₂ into a single risk model required understanding complex ecological interactions. Resolved through a targeted literature review grounding the model in peer-reviewed science.

Cross-Functional Data Handoff: Coordinating outputs across four downstream teams with different format requirements. Resolved through clear documentation, multiple export formats, and early delivery of test datasets.

Learning Outcomes

Data Quality is Foundational: Data quality directly impacts every downstream module. Careful, thoughtful data work prevents costly errors later in the pipeline.

Communication is Critical: Working across eight specialisations required clear documentation, early coordination, and the ability to explain technical decisions to non-specialist teammates.

Science Matters: Grounding engineering decisions in peer-reviewed literature and scientific first principles led to more robust, trustworthy outputs.

Iteration is Essential: The best solutions emerged through testing, validation, feedback, and refinement — not from getting everything right the first time.

Real-World Impact: Building tools that can help NGOs monitor ocean health and respond to environmental crises made the technical work deeply meaningful.

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